

Genomic function features of colorectal cancer-associated bacteria compared to colorectal health-associated bacteria

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Introduction

Colorectal cancer is one of the most common human cancers around the world with two million new patients every year¹. Gut microbiome is proved an important factor in the tumor development^{2,3}. Research on the gut microbiome is significant in preventing and conquering colorectal cancer. At current, several colorectal tumor-associated bacteria species (*Bacteroides fragilis*, *Fusobacterium nucleatum*, *parvimonas spp.*, *prevotella spp.* etc.) in gut microbiome have been identified^{4,5}. Previous research found different mechanisms of these bacteria to colorectal tumors, such as the secretion of microRNA, genotoxins, and specific metabolites⁶⁻⁸. However, these studies failed to evaluate the common characteristics of this group of bacteria compared to probiotics. Due to the lack of research in this field, the current study aims to verify whether the colorectal tumor-associated bacteria have genomic function features to the colorectal health-associated bacteria. with a hypothesis that this group of bacteria has genomic function features to the healthy bacteria.

Materials and Method

Materials

Genomes of seven colorectal cancer associated (CRC) bacteria species and seven colorectal health associated (CRH) bacteria species (selected based on

previous research^{4,9}) from the National Center for Biotechnology Information (NCBI) database. The genome information was listed in Table.1.

Method

Two groups of bacteria (Group A: CRC bacteria, Group B: CRH bacteria) were selected for comparison. The sample genomes were first checked by Quast for quality evaluation. Then, the genes of each genome were predicted by gms2.pl software with gmhmmp2 keys. The gene prediction results were compared by numbers of gene encoded proteins (faa form) with the sample references from NCBI database for checkage. The genes were next functionally annotated by kofamscan software using KEGG database. The gene annotation results were checked by KEGG mapper function by generated ko-list. The number of encoded enzymes for each sample in six KEGG metabolism pathways (pathways selected by excluding central cell life metabolism pathways and few encoded enzymes (<3) pathways) were compared as genomic function comparison. The functional differences were analyzed by multiple t-test. Visualization of results was performed on R software (4.3 version) with ggplot2, tidyverse, pheatmap, RColorBrewer software packages.

	Sample	Bacteria Species	Reference Genome	RefSeq
Group A Colorectal cancer associated bacteria	1	<i>Fusobacterium nucleatum</i>	<i>Fusobacterium nucleatum</i> subsp. <i>Nucleatum</i>	NZ_CP028101.1
	2	<i>Prevotella intermedia</i>	<i>Prevotella intermedia</i> ATCC 25611	NZ_CP019300.1/NZ_CP019301.1
	3	<i>Prevotella bivia</i>	<i>Prevotella bivia</i> DSM 20514	NZ_AJVZ00000000.1
	4	<i>Thermanaerovibrio acidaminovorans</i>	<i>Thermanaerovibrio acidaminovorans</i> DSM 6589	NC_013522.1
	5	<i>Porphyromonas gingivalis</i>	<i>Porphyromonas gingivalis</i> ATCC 33277	NC_010729.1
	6	<i>Helicobacter pylori</i>	<i>Helicobacter pylori</i>	NZ_CP071982.1/NZ_CP071983.1
	7	<i>Campylobacter jejuni</i>	<i>Campylobacter jejuni</i> subsp. <i>jejuni</i> NCTC 11168 = ATCC 700819	NC_002163.1
Group B Colorectal health associated bacteria	8	<i>Bifidobacterium longum</i>	<i>Bifidobacterium longum</i> subsp. <i>longum</i> JCM 1217	NC_015067.1
	9	<i>Bifidobacterium porcinum</i>	<i>Bifidobacterium porcinum</i> DSM 17755	NZ_JDTQ00000000.1
	10	<i>Lactobacillus acidophilus</i>	<i>Lactobacillus acidophilus</i> La-14	NC_021181.2
	11	<i>Lactobacillus crispatus</i>	<i>Lactobacillus crispatus</i>	NZ_CP039266.1/NZ_CP039267.1
	12	<i>Roseburia lenta</i>	<i>Roseburia lenta</i>	NZ_JACOPG000000000.1
	13	<i>Ruminococcus bovis</i>	<i>Ruminococcus bovis</i>	NZ_CP039381.1
	14	<i>Eubacterium coprostanoligenes</i>	<i>Eubacterium coprostanoligenes</i>	NZ_FUWW000000000.1

Table.1 The basic genome information of 14 samples

Results

Genome evaluation and Gene prediction Summary

In data preparation process, the genome of 14 samples were evaluated first to ensure the quality. Evaluation results of each sample were listed in Table.2. In general, the sample genomes were high-standard and integrated. Most of the genomes were at complete genome assembly levels while few were at contig levels. All the genome assemblies were evaluated as high N50 values (Represented the length of contig which was more than half of the total assembly length after accumulation. Under same genome size condition, higher N50 value indicated better integrity of the assembly) and low L50 values (Represented the order of the N50 contig).

Genes of sample genomes were then predicted. The prediction results were compared to the references for checkage by comparing the numbers of gene encoded proteins (Fig.1). The number of the proteins in most sample genomes was consistent with the reference with few pseudogenes, showing the accuracy of the gene prediction. Particularly, the gene numbers of all samples were relatively closed to each other as around 2000, excluding the influence of genome size on the comparison results.

Different Correlation of Genomic Function to six Metabolic Pathways Between Two Bacteria Groups

Predicted genes were next functional annotated. Genomic function difference of CRC bacteria group and CRH bacteria group was compared. The difference of sample genomic function could be compared

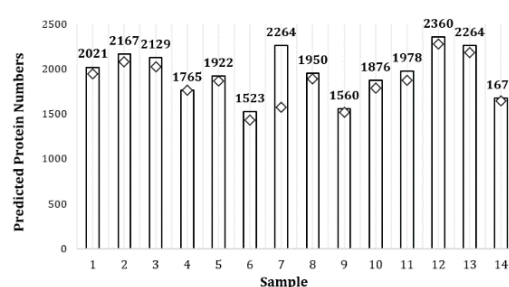


Fig.1 The gene prediction results compared to the reference.

through the varied quantities of encoded enzymes (each encoded enzyme participated in one biochemical reaction of the pathway, representing the bacteria function) in multiple KEGG metabolic pathways. From the results, CRC bacteria (Group A) encoded enzymes in the six pathways were significantly different from CRH bacteria (Group B) with p values less than 0.05 (Fig.2 A and B). In addition, the clustering heat map showed the correlation of genomic functions for different sample species to six metabolic pathways. Generally, excluded individual cases, the genomic function of CRC bacteria would be positively correlated to riboflavin metabolism, folate biosynthesis, carbon metabolism, and biosynthesis of cofactors pathways. These pathways mainly functioned as the synthesis of the cofactors and the B vitamins, and the metabolism of energy. The genomic functions of CRC bacteria were also negatively correlated to human absorbable sugar metabolism pathways, which included galactose metabolism, and starch and sucrose metabolism pathway. The correlation of CRC bacteria group to six pathways was opposite from CRH bacteria group.

Sample	Bacteria Species	Assembly level	Total length	N50	L50	Contig
1	<i>Fusobacterium nucleatum</i>	Complete genome	2180101	2180101	1	1
2	<i>Prevotella intermedia</i>	Complete genome	2674170	1965935	1	2
3	<i>Prevotella bivia</i>	Scaffold	2521238	1982184	1	3
4	<i>Thermanaerovibrio acidaminovorans</i>	Complete genome	1848474	1848474	1	1
5	<i>Porphyromonas gingivalis</i>	Complete genome	2354886	2354886	1	1
6	<i>Helicobacter pylori</i>	Complete genome	1624458	1615199	2	2
7	<i>Campylobacter jejuni</i>	Complete genome	1641481	1641481	1	1
8	<i>Bifidobacterium longum</i>	Complete genome	2385164	2385164	1	1
9	<i>Bifidobacterium porcinum</i>	Contig	2060411	152447	4	28
10	<i>Lactobacillus acidophilus</i>	Complete genome	1991579	1991579	1	1
11	<i>Lactobacillus crispatus</i>	Complete genome	2039404	2023061	2	2
12	<i>Roseburia lenta</i>	Contig	2558168	738110	2	10
13	<i>Ruminococcus bovis</i>	Complete genome	2440231	2440231	1	1
14	<i>Eubacterium coprostanoligenes</i>	Contig	1823434	91496	7	36

Table.2 The genome evaluation results of 14 samples.

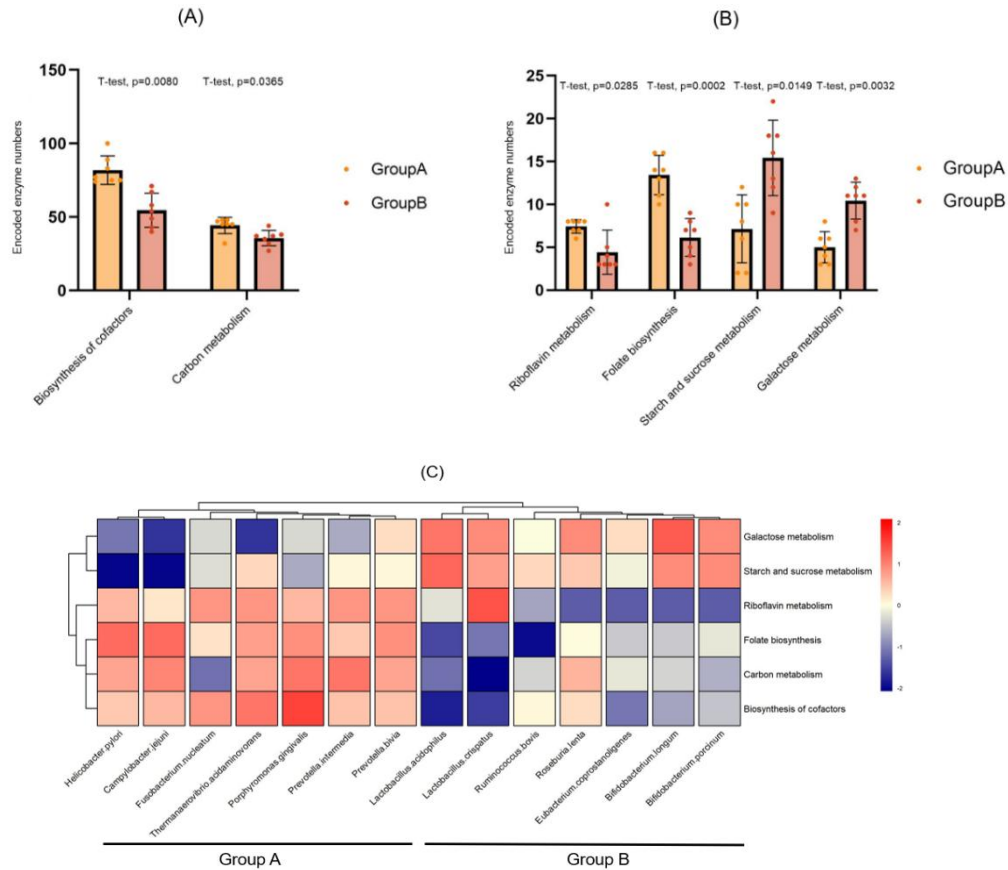


Fig.2 A. Numbers of encoded enzyme comparison within groups with mean comparison P values in abundant encoded enzyme pathways. B. Numbers of encoded enzyme comparison within groups with mean comparison P values in relatively low encoded enzyme pathways. C. Clustering heat map of bacteria genomic functions to KEGG metabolic pathways based on the quantity of encoded enzymes. The map showed the correlations between two variables. Red color represented positive correlation while blue color represented negative correlation.

Discussion and Conclusion

In this study, colorectal cancer-associated bacteria and colorectal health-associated bacteria were performed comparative genomes with the aim of screening characteristic of CRC bacteria genomic function compared to probiotics. The hypothesis that the colorectal tumor-associated bacteria had specific genomic function features that differed from the colorectal health-associated bacteria was supported by the results.

The genome evaluation and gene prediction results confirmed the integrity of genome samples and the accuracy of gene prediction, respectively. Similar gene

numbers between samples could reduce the bias of the genome size on function comparison outcomes, increasing the research validity. The genomic function comparison results revealed that the functions of CRC microbes would be more correlated to the cofactor, B vitamin, and energy metabolism pathways, and less to human absorbable sugar metabolism pathways, contrasted from healthy microbes.

These findings are consistent with previous research. On the one side, CRC bacteria is proved the capacity to interference with the host's metabolism, establishing interactions between the colorectal tumor cells and the immune systems⁴⁻⁶. Cofactors and B vitamins mainly functioned in regulating these processes. On the other side, the secretion of carbohydrate enzymes for CRH bacteria is found to positively influence the gut

microbiome's metabolic functions, prompting colorectal health^{9,10}. Therefore, CRC bacteria's negative correlation of genomic function to these pathways was rational.

Due to the limitations in this study, the relation of CRC bacteria genomic function features to colorectal cancer is still required for further analysis. This study finds a distinguished genomic function correlation mode of CRC bacteria in six metabolic pathways compared to CRH bacteria, which could provide directions for future CRC bacteria species discovery.

In conclusion, the study compared the genomic function between the colorectal cancer-associated bacteria and the colorectal health-associated bacteria to verify whether the cancer-related bacteria had genomic function features compared to healthy bacteria. The hypothesis is confirmed by the results that the genomic function of cancer related bacteria is positively correlated to cofactors, B vitamin, and energy metabolism pathways while negatively correlated to human absorbable sugar metabolism pathways. The research findings could contribute to the identification of colorectal cancer-related bacteria species in the future.

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